
Should a police force let a model decide who gets searched?

Interpretability, stability and algorithmic fairness on 58,865 discretionary traffic-stop searches

Nashville Metropolitan PD, 2010–2018 · Stanford Open Policing Project

HEC Paris · MSc Data Science & AI for Business · Prof. Christophe Pérignon · 28 September 2026

The client's decision, not ours

A police force has finite search capacity.

Roughly 2,000 discretionary searches in our test window. Which stops should get one?

The offer on the table: rank every stop by predicted probability of contraband, then search the top K.

That is a scoring model on a binary target — exactly what the brief asks for.

Three model families, four dimensions.

White box, machine learning, tabular foundation model — each judged on predictive performance (statistical AND economic), interpretability, stability and fairness.

Our answer: build all three, then recommend deploying none of them — and the reason is not that they score badly. It is what they are scoring.

Nashville, because the outcome is observed

3,078,116

traffic stops, 2010–2018

127,122

ended in a search

58,865

consent searches — our sample

9,113

test searches, 2016+

Stanford Open Policing Project — Pierson et al., Nature Human Behaviour 4 (2020).

Open Data Commons Attribution Licence. Cited in the repository, in the app and here.

Why Nashville: `contraband_found` is recorded for every search.

Most forces record that a search happened, not what it produced. No outcome, no target and no outcome test.

Temporal split at 2016 — train before, test after.

Not random: a random split leaks the future into the past through the same officer, the same shift and the same policy regime.

What we predict — and the trap underneath it

$Y = 1$ if contraband was found. $\hat{Y} = 1$ if the model would search. D = driver race.

Note the inversion against the course convention: here $Y = 1$ is the outcome that JUSTIFIES a search, so the favourable outcome for the individual is NOT being searched. We report every fairness metric under both codings.

Selective labels: we observe contraband only for stops an officer already chose to search.

Every model here is trained on the officers' own decisions. It learns to predict the result of a filter we cannot see through — not the underlying rate of contraband.

So the honest question is not “how accurate is it?”

It is: what does this score reproduce? That reframing drives the whole analysis.

A model trained on who officers chose to search can only ever be as fair as the choices it is imitating.

Exactly which searches are in our sample

69,251

flagged consent

58,865

consent AND no other authority

10,386

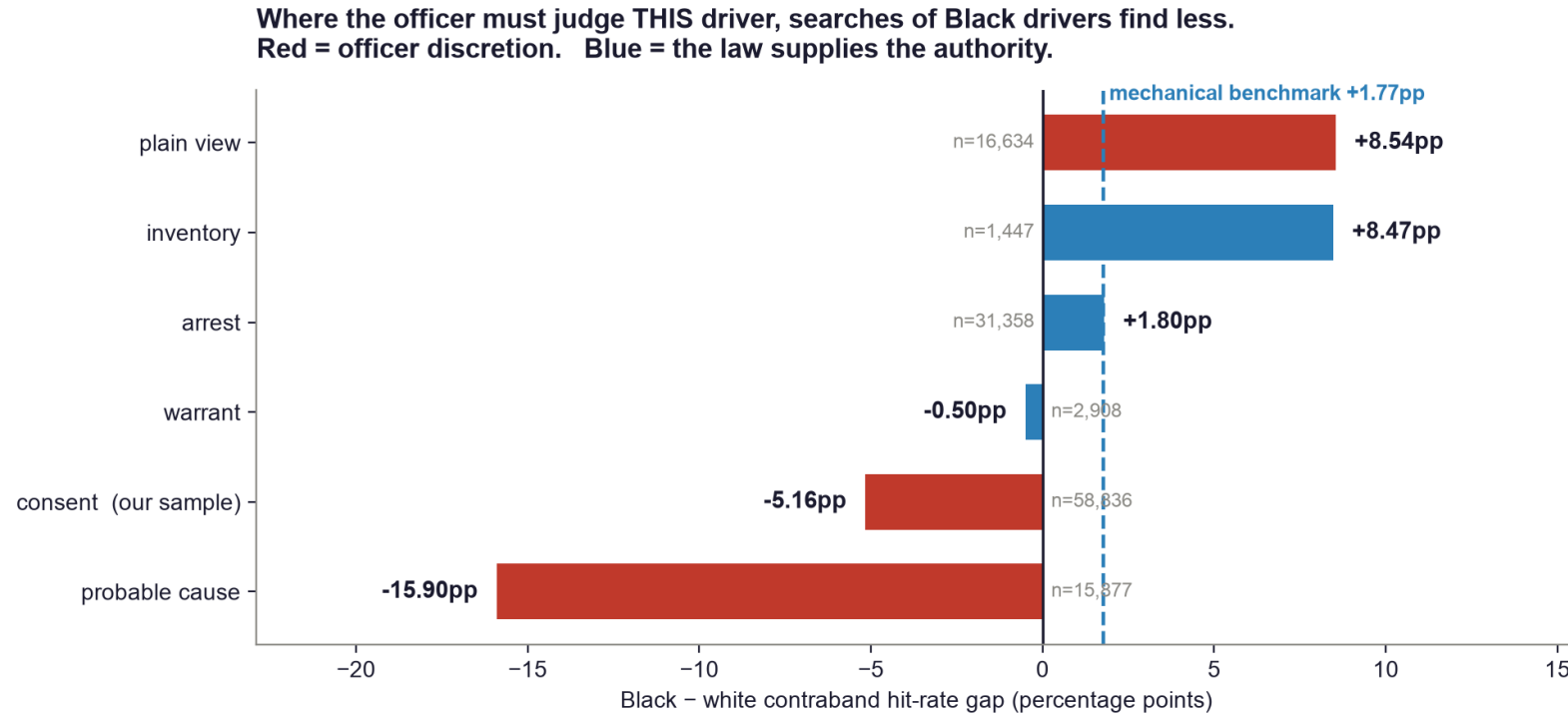
excluded — also carry arrest,
warrant, inventory or plain view

The 10,386 excluded searches are not neutral: they hit at 28.7% against 16.9% for the ones we keep, and 76% of them ended in arrest.

The exclusion is on the legal basis, not on the outcome — but that basis is outcome-correlated, so the purification is not free. We therefore report both.

Black–white hit-rate gap: -5.16pp on the pure stratum · -4.60pp on all 69,251 consent-flagged searches. Same sign, same conclusion.

The outcome test, by legal basis



Consent -5.16pp and probable cause -15.90pp — the two bases needing the officer's judgement about THIS driver — are both negative. The mechanical bases sit at +1.77pp.

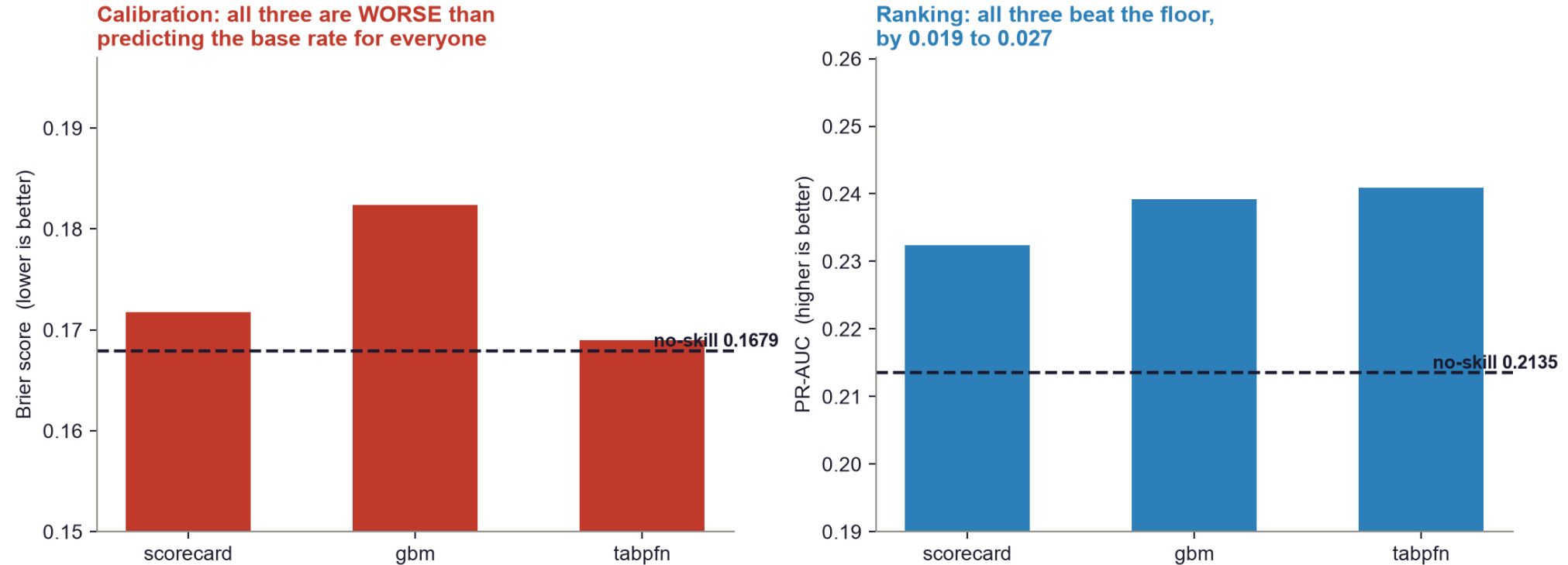
Three families, one matched comparison

	White box	Machine learning	Tabular foundation
Model	Logistic scorecard	XGBoost	TabPFN v2
Explanation	the coefficients ARE the model	TreeSHAP (exact)	occlusion (approximate)
Train rows	2,000	2,000	2,000
AUC	0.5329	0.5478	0.5529
Brier	0.1718	0.1824	0.1690

Test set: the same 9,113 searches from 2016 onward for every arm.

All three train on the identical 2,000 rows — TabPFN's context ceiling. Anything else compares training budgets, not model families.

Measured against doing nothing



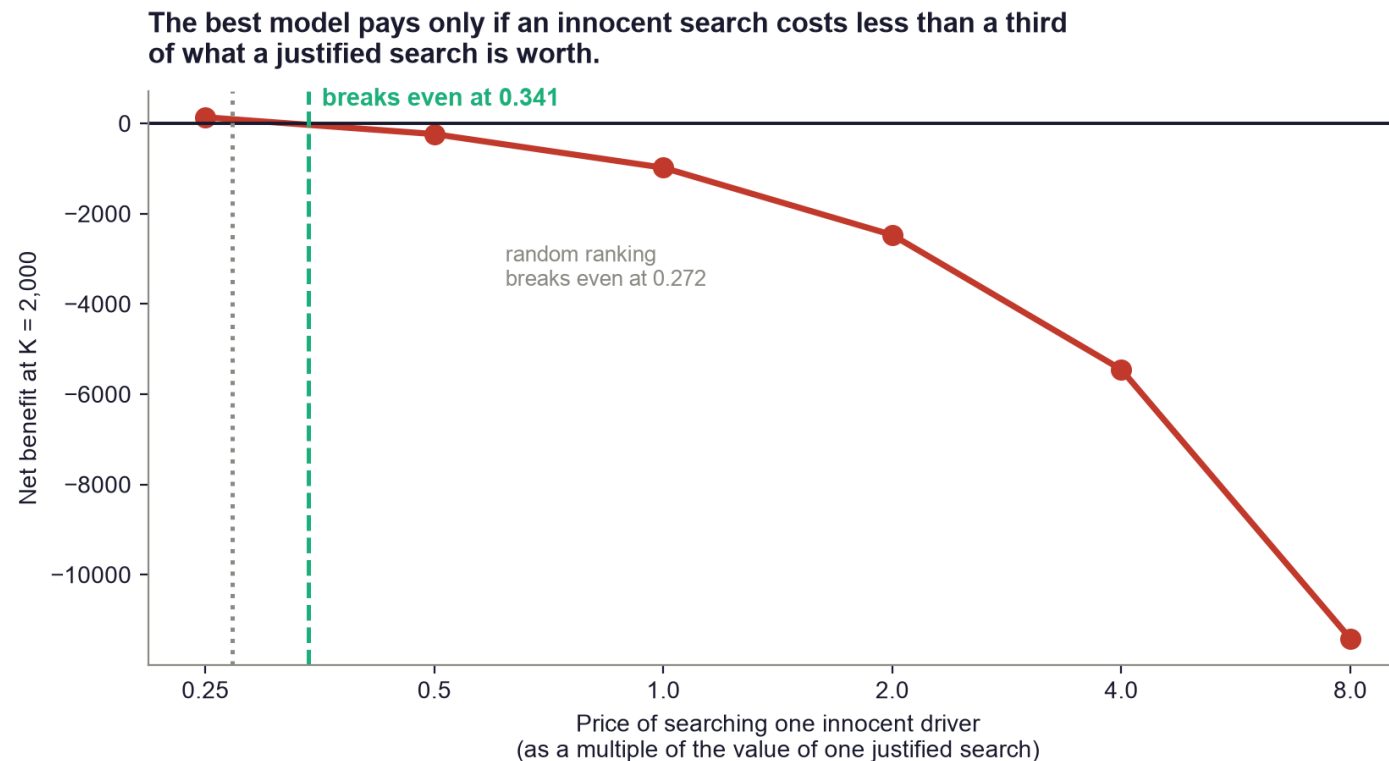
Not one of the three beats the Brier floor of 0.1679 — the score you get by predicting the base rate 21.4% for every single driver.

Are the three actually different? No.

Comparison	Δ AUC	95% CI	p	
XGBoost vs TabPFN	-0.0026	[-0.0157, +0.0104]	0.691	not significant
XGBoost vs tree	+0.0062	[-0.0091, +0.0215]	0.431	not significant
Tree vs TabPFN	-0.0088	[-0.0212, +0.0036]	0.163	not significant
Scorecard vs XGBoost	-0.0173	[-0.0317, -0.0028]	0.019	SIGNIFICANT
Scorecard vs TabPFN	-0.0199	[-0.0291, -0.0108]	<0.001	SIGNIFICANT

DeLong paired test on correlated AUCs. The two black boxes are statistically indistinguishable, so we do not rank them — any choice between them must be made on the other three dimensions.

What it would be worth to deploy



The best arm breaks even only while an innocent driver's search costs less than 0.341 of a justified one. Random ranking breaks even at 0.272 — the model buys 0.070 of headroom.

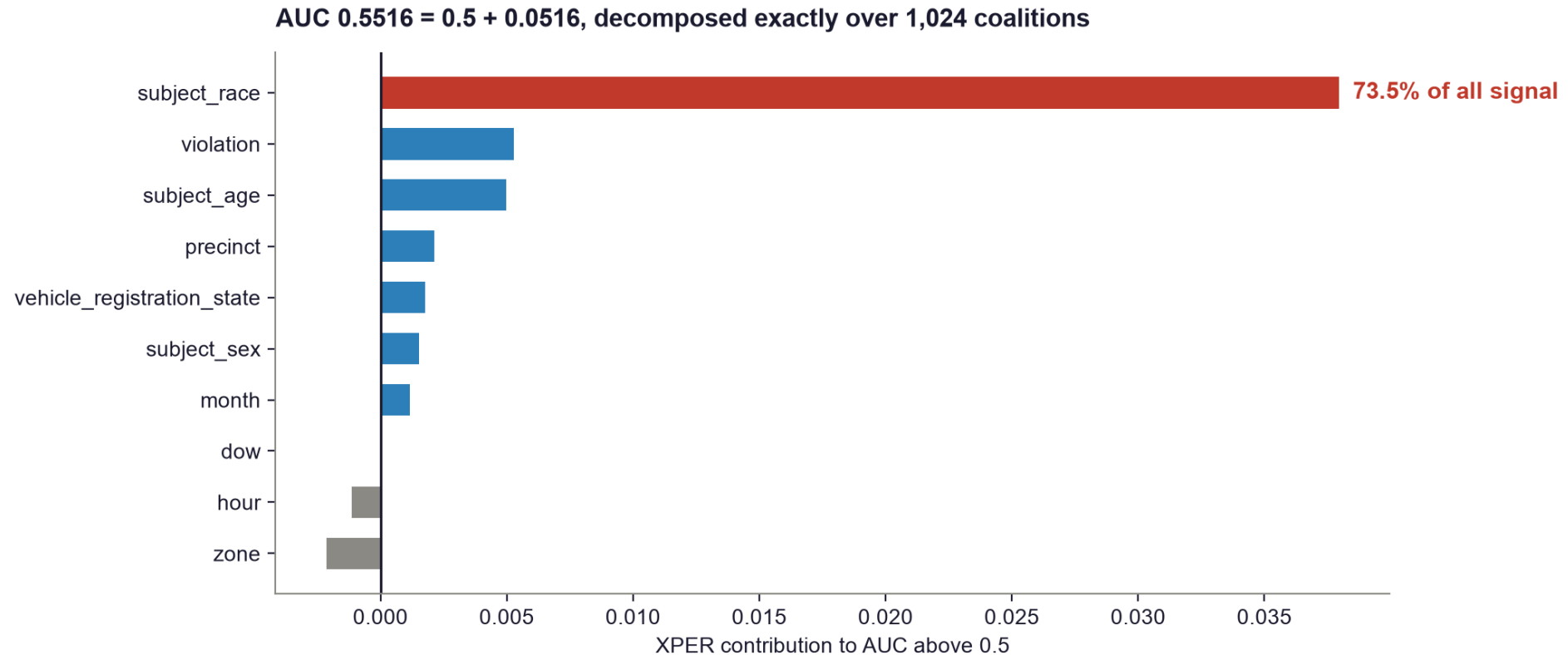
Three models, three grades of explanation

Arm	Local explanation	Exactness	What it costs
Scorecard	Shapley in closed form: $\text{coef} \times (x - E[x])$	EXACT	nothing — it is algebra
XGBoost	TreeSHAP, native to the model	EXACT	nothing
TabPFN	occlusion against a background value	APPROXIMATE	ignores interactions; error cannot be bounded

The foundation model has no native attribution at all.

Not a gap in our implementation — TabPFN exposes none. For a decision that must be justified to the person searched, an unbounded approximation is a real cost, not a technical footnote.

Where the signal actually comes from



Driver race alone accounts for 73.5% of everything the white-box model knows above chance. Exact decomposition over 1,024 coalitions; efficiency gap 3×10^{-17} .

The score does not survive its own assumptions

0.6403

pooled AUC, best model

0.5597

the same model, WITHIN officer

~80%

of the gain is between officers,
not inside them

Leave-one-officer-out: the model largely learns WHICH OFFICER made the stop.

Deployed, it would rank a new officer's stops with almost no signal. The pooled number is the one you would quote; the within-officer number is the one you would live with.

Temporal split plus an officer-clustered bootstrap over 1,477 officers.

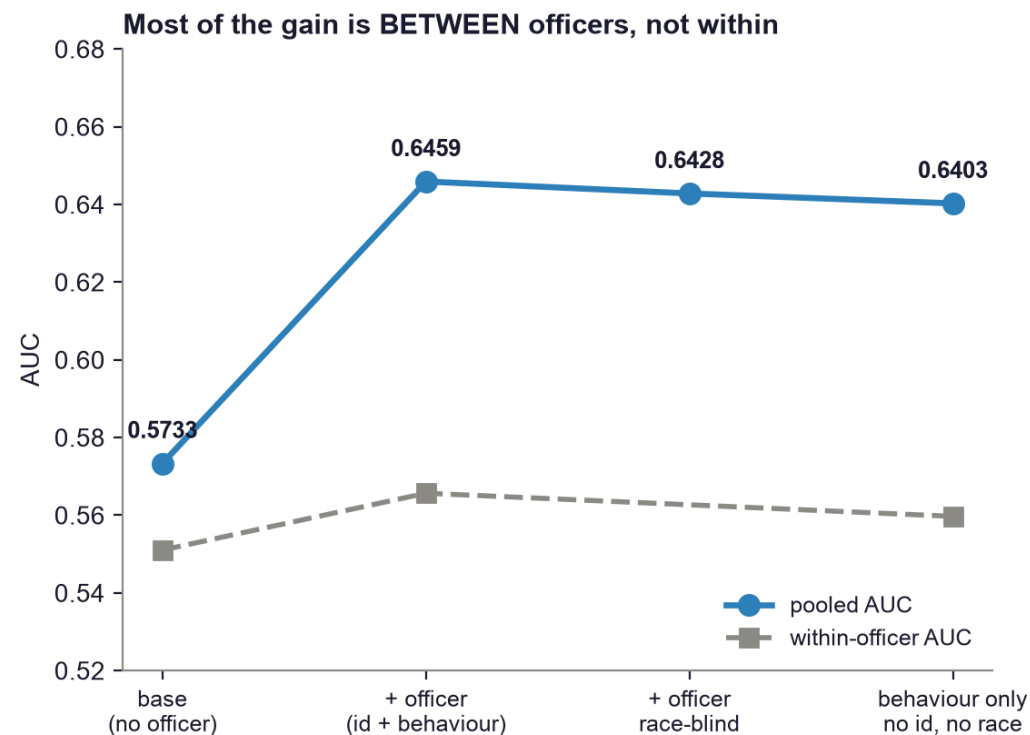
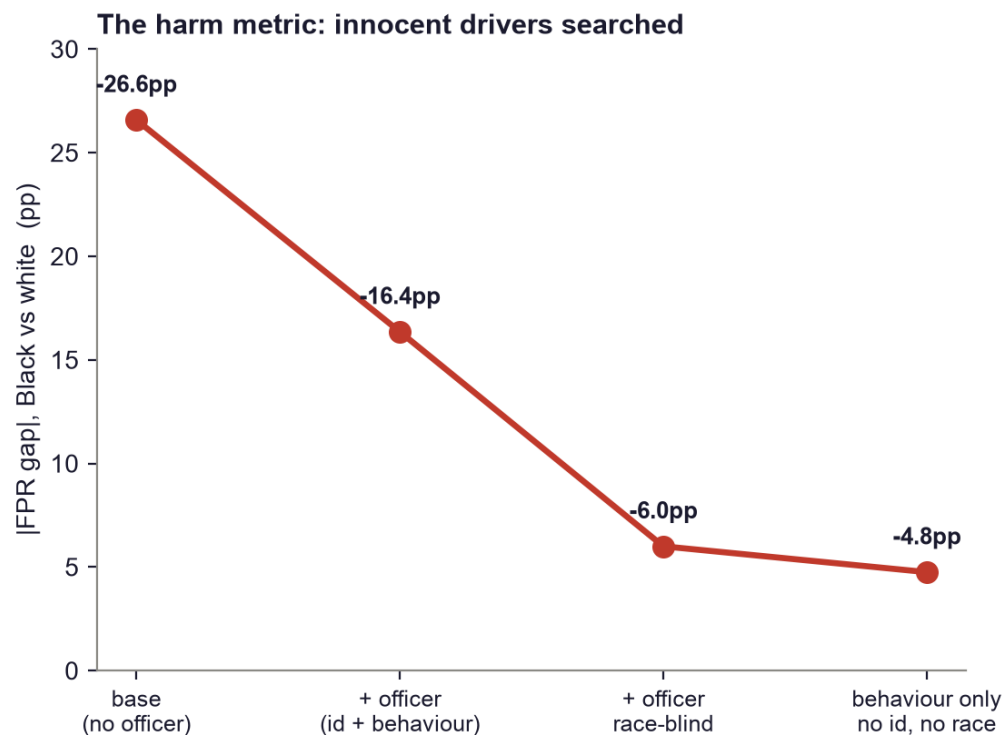
The officer block gains +0.072 AUC, CI [+0.052, +0.095] — the only performance claim in this project an order of magnitude clear of the noise.

The model fails independence — backwards

Criterion	Test	Result on the base model
Independence	$\hat{Y} \perp D$ — statistical parity	FAILS: $\chi^2 = 863$, selection gap -27.9pp
Separation	equal FPR / TPR across groups	FAILS: FPR gap -26.6pp — the harm metric
Sufficiency	equal hit rate at equal score	closest to holding; calibration by race is in the app

Read the SIGN. The model searches Black drivers LESS than white drivers (-27.9pp) — it inverts real officer behaviour, having learned that consent searches of Black drivers find less. A disparity in the “favourable” direction is still a disparity, and it is still race doing the work.

Give it the officer, and race stops mattering



Removing race entirely costs 0.0056 AUC ($p = 0.144$, not significant) and takes the FPR gap from -16.4pp to -4.8pp. Going race-blind is statistically free.

But nothing in the data repairs parity

0 of 32

features whose adjustment
restores statistical parity

65

χ^2 of the best model —
still rejects independence

23

best single repair
(off_consent_hit_rate_365d_shrunk),
still far above 3.84

Fairness Partial Dependence Plots sweep every feature and ask: if we intervened on this one, would the disparity close?

No, for all 32. The disparity is not carried by any single removable column — it is spread across the feature set, which is what proxy discrimination looks like when you go looking for it.

This is why “just drop the sensitive feature” is not a fix.

We dropped race and the disparity persisted at −4.8pp. There is no column to delete.

What we are not claiming

Selective labels. We never observe contraband for stops that were not searched.

Every number here is conditional on an officer having already decided to search.

No validation set. Tree depth, context size, feature blocks and K were all chosen by reading the same 9,113-row test set that produces every reported number.

Treat any AUC difference below 0.01 as inside the noise. Only the officer block (+0.072, $p = 6.6e^{-23}$) is clear of it.

Inframarginality. A hit-rate gap is not proof of a different threshold.

The outcome test is suggestive, not dispositive; a full threshold test is the next step.

Provenance. The officer-model runs were done interactively and only their outputs were committed — the driver script is not in the repository.

We say this before being asked. The reproducible path today is `scripts_claude/evaluate_officer_models.py`, which replicates the design and gives 0.6397.

Deploy none of the three

Not because they score badly — because of what they score.

73% of the white box's signal is race. The best model's edge is mostly which officer made the stop. None of the three beats a constant prediction on calibration. Net benefit is negative at any defensible price for searching an innocent driver.

If a model must ship: the scorecard, race-blind, with officer behaviour features.

It costs 0.018 AUC against the black boxes — the only statistically significant performance difference we found — and buys an exact, auditable explanation for every individual decision. That is the trustworthy-AI trade the brief asks us to make.

The finding worth more than the model: the outcome test itself.

Consent -5.16pp, probable cause -15.90pp. Deploy that as an officer-level audit, not a driver-level score. It needs no model, it has a victim you can name, and it points at the decision that is actually going wrong.

A trustworthy system here is one that audits the officers, not one that ranks the drivers.

Backup - the comparison group, decomposed

Legal basis	n	white hit	Black hit	gap	discretionary?
probable cause	15,877	61.21%	45.31%	-15.90pp	YES
consent (our sample)	58,836	20.84%	15.68%	-5.16pp	YES
warrant	2,908	14.99%	14.50%	-0.50pp	no
arrest	31,358	19.42%	21.21%	+1.80pp	no
inventory	1,447	14.98%	23.44%	+8.47pp	no
plain view	16,634	5.91%	14.44%	+8.54pp	YES

Mechanical (arrest + warrant + inventory): n = 35,713, gap +1.77pp. Still discretionary (probable cause + plain view): n = 32,511, gap +11.17pp — 47.7% of the comparison group. Pooled over everything: +0.82pp. n is the count each gap is computed on: drivers recorded white, Black or Hispanic.

Backup · every arm against its no-skill floor

Arm	AUC	PR-AUC	vs PR floor	Brier	vs Brier floor
scorecard	0.5329	0.2324	+0.0189	0.1718	+0.0038 worse
gbm	0.5478	0.2392	+0.0257	0.1824	+0.0144 worse
tabpfn	0.5529	0.2409	+0.0273	0.1690	+0.0010 worse

Floors from a constant prediction at the test base rate 0.2135: Brier 0.1679, PR-AUC 0.2135. At $K = 2,000$: TabPFN 509 hits against 427 for random ranking.

Backup · the fairness ladder in numbers

Model	cols	AUC	within-officer	sel gap	FPR gap	χ^2	TOST δ^*
base (no officer)	17	0.5733	0.5510	-27.9pp	-26.6pp	863	0.30
+ officer (id + behaviour)	42	0.6459	0.5657	-20.3pp	-16.4pp	463	0.22
+ officer race-blind	40	0.6428	—	-9.4pp	-6.0pp	100	0.11
behaviour only no id, no race	32	0.6403	0.5597	-7.5pp	-4.8pp	65	0.10

TOST δ^ is the smallest equivalence margin at which the model could be certified fair. It falls from 0.30 to 0.10 — an improvement, and still a 10-point selection gap you would have to declare acceptable.*